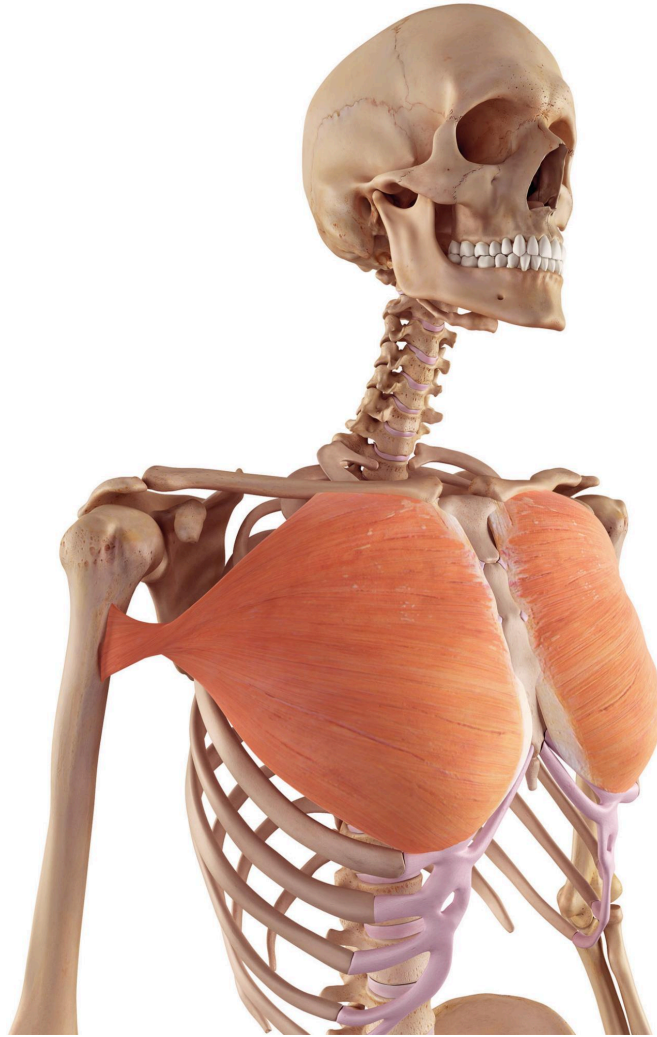


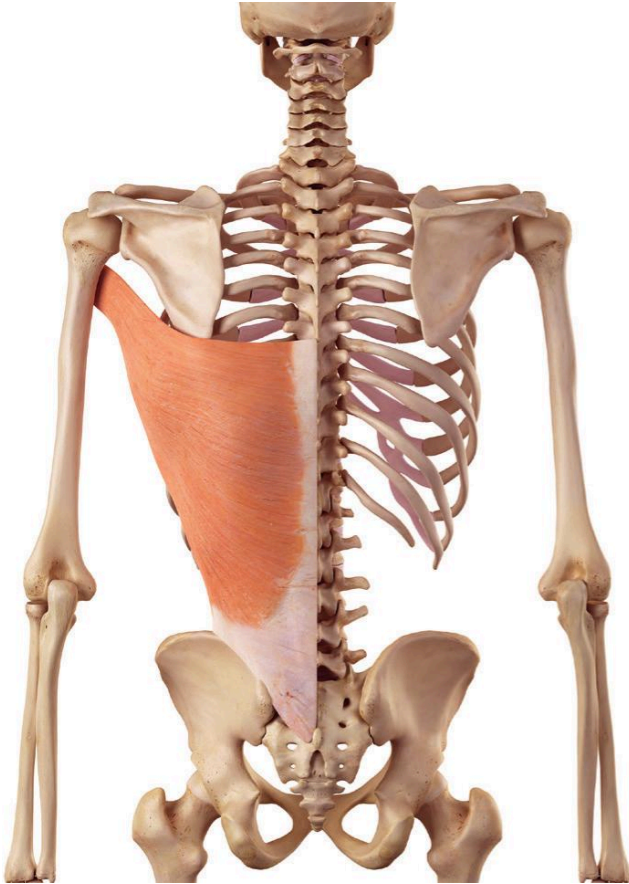
BRIEF NOTES ON ANATOMY



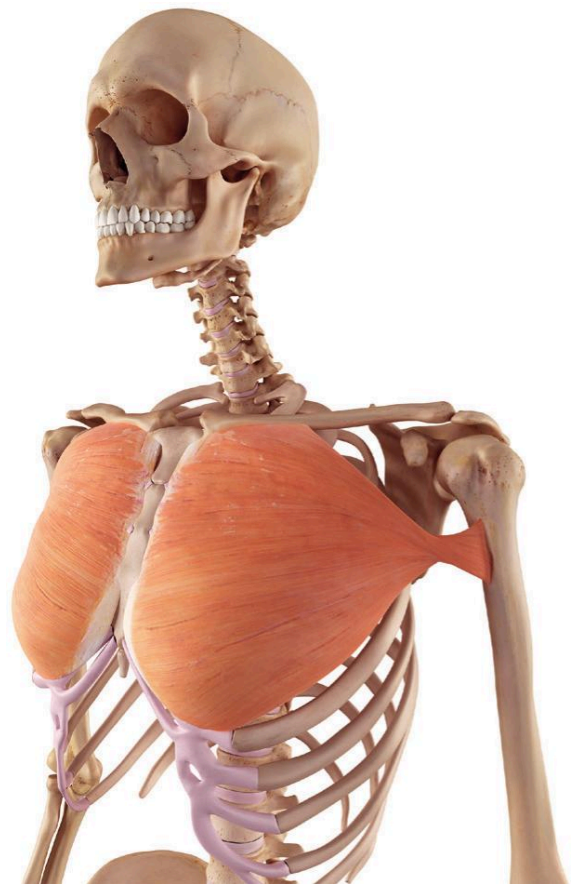
You might be wondering why there's a section on anatomy in a book about training for climbing. However, understanding how our body is designed will give us a better insight into how it works. One of the objectives of this book is to understand how our body works so we can understand how to design a training plan, why an exercise has to be done in a certain way or why choosing a different exercise could lead to injury.

To simplify the information and make it easier to digest, let's say that most of the time in climbing, the muscles in the upper body are used for *pulling* (the basic action of holding on and pulling up), while the muscles in the lower body are used for *pushing*, or propelling, us up the wall. Between the upper and the lower body lies the core (a firm favourite at the gym), linking these parts of the body and bringing both actions together into one fluid upward motion.

This chapter looks at which muscles and anatomical structures are most important for climbing, keeping the level of detail (and tedium) to a minimum and drawing on information provided by Schünke *et al.* (2005).



LATISSIMUS DORSI



PECTORALIS MAJOR

UPPER BODY: PULLING MUSCLES

To help understand the anatomy of the upper body, the muscles can be divided into three main groups:

- *Torso muscles*: the latissimus dorsi and the pectoralis major are the main muscles in this group and the most important for climbing. The latissimus dorsi is the number one pulling muscle in climbing, while the pectoralis major, far from being its antagonist, helps to stabilise the glenohumeral joint (shoulder) and to move the arm towards and away from the body.



TRAPEZIUS



RHOMBOIDS



LEFT: SERRATUS ANTERIOR (NOTE THE EXTENSION FROM THE SCAPULA ROUND TO THE RIBS)



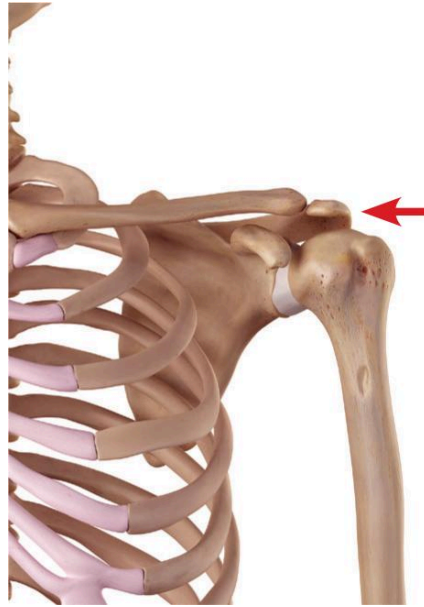
CENTRE: DELTOIDS



RIGHT: NOTE THE EXTENSION OF THE DELTOIDS FROM THE CLAVICLE TO THE POSTERIOR PART AT THE SCAPULA



ROTATOR CUFF



SUBACROMIAL SPACE



SUPRASPINATUS MUSCLE IN THE
SUBACROMIAL SPACE

- *Shoulder girdle*: in addition to the shoulder joint, the arm is also attached to the torso by a group of joints and muscles (a girdle) that connects the humerus to the scapula and the clavicle. It also connects the scapula to the spine and the rib cage, and the clavicle to the sternum. This aside, let's look at the shoulder girdle in a little more depth. By understanding the scope and complexity of this structure, we'll see why its stability is so important. On the one hand, the large latissimus dorsi, pectoralis major, serratus anterior and the anterior deltoid, helped by other smaller muscles like the teres major and the subscapularis, are responsible for internal rotation and scapular protraction. In contrast, only the infraspinatus muscle and the teres minor are responsible for external rotation, and only the trapezius and rhomboids are responsible for scapular retraction. As a result, if we don't actively counter the predominant action (internal rotation and scapular protraction), we'll end up not only with rounded shoulders but at a much higher risk of injury.

On the other hand, the socket of the shoulder joint (where the large head of the humerus articulates with the glenoid fossa of the scapula) is so shallow that the glenoid ligament, a fibrocartilaginous structure, is needed to deepen the socket and improve contact. The difference in size between the humeral head and the glenoid fossa makes the shoulder joint incredibly mobile but also very unstable, leading to a high risk of injury. On the subject of injury, the subacromial space also deserves a mention. This space is located between the humerus and the acromion (see image, [above](#)), and the supraspinatus tendon runs through it. If the scapula is not properly stabilised (for example, retracted and depressed for deadhangs), this space is compressed and the supraspinatus tendon can become impinged. Bursitis, tendinosis and even tears are some of the nasty injuries that can occur (not to mention long periods of intense rehabilitation and retraining) if we neglect the shoulder girdle and the subacromial space.



BICEPS BRACHII



TRICEPS BRACHII (NOTE THE LONG HEAD THAT RISES FROM THE SCAPULA)

- *Upper arm and forearm muscles:* despite being part of the most important limb in climbing, the upper arm doesn't really have that many muscles. It contains the biceps and the brachialis, which are elbow flexors, and the triceps, which is an extensor muscle. The long head of the triceps also helps the latissimus dorsi to move the humerus towards the torso.

The forearm and the hand, the stars of the show, are a little more complex. There are 19 muscles in the forearm (eight flexors, eight extensors and three radial) and 11 in the hand. In climbing, we need a good degree of balance and synergy between all these muscles and a high level of strength in different grip positions. That said, the flexor digitorum profundus (FDP) is *the* most important muscle in climbing (Philippe *et al.*, 2012). This is because it flexes the interphalangeal joints in digits two to five, which are the fingers used for dragging. In turn, the flexor carpi ulnaris and the palmaris longus muscle are used most with slopers. The extensors help maintain joint stability, and they move the wrist and forearm into the right position for gripping and pulling. As for pinches, the muscles in the hand do most of the work.



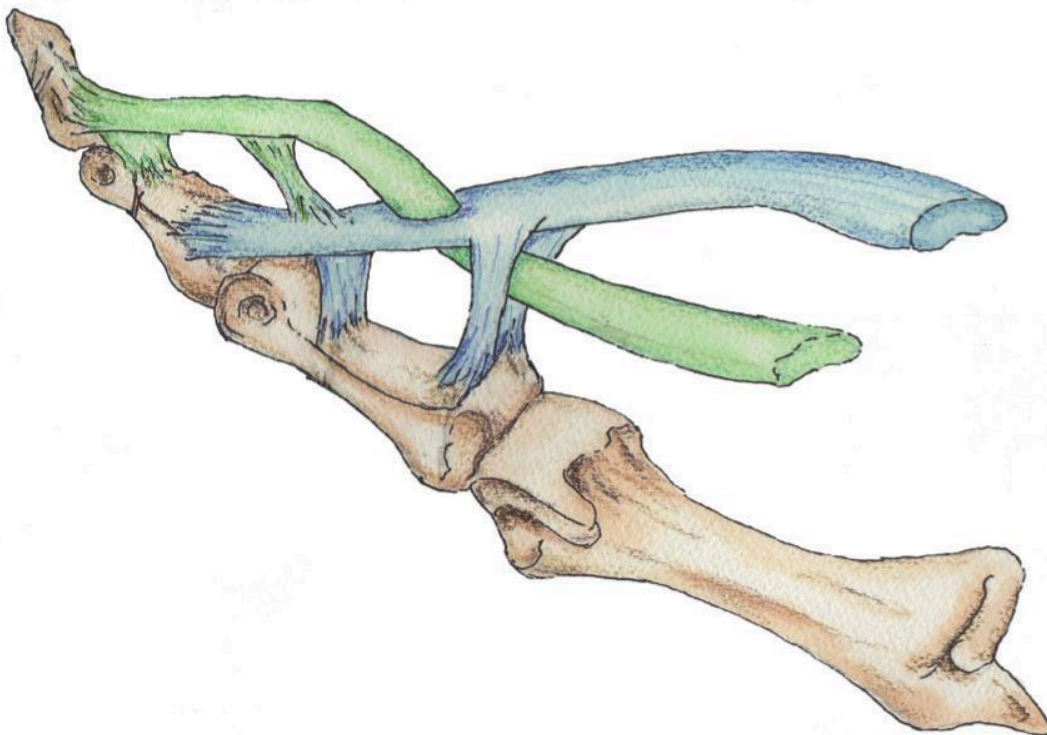
FOREARM RADIAL MUSCLES



FOREARM FLEXOR MUSCLES

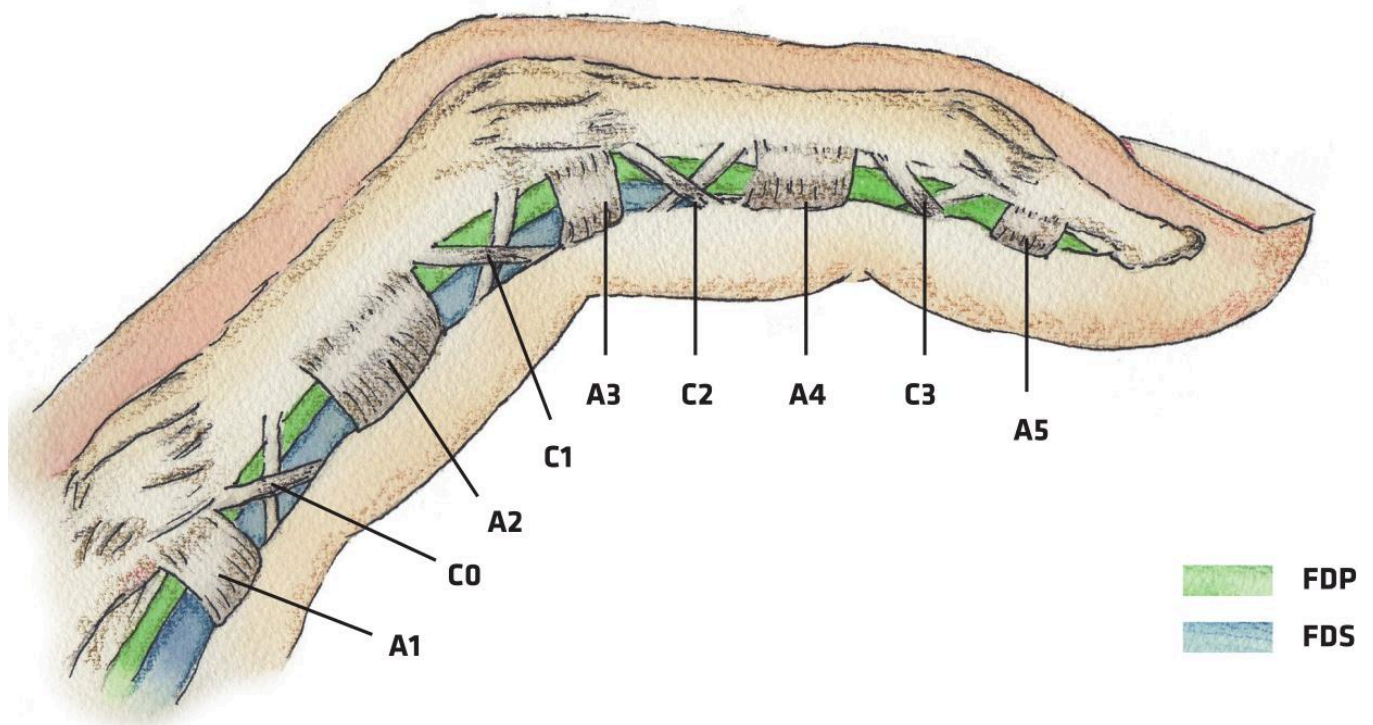


FOREARM EXTENSOR MUSCLES



FDP
FDS

INSERTION IN THE FINGER BONES OF THE FLEXOR DIGITORUM SUPERFICIALIS (FDS) AND THE FLEXOR DIGITORUM PROFUNDUS (FDP)



ANNULAR PULLEYS (A) AND CRUCIATE PULLEYS (C), WHICH GUIDE AND SUPPORT THE FDS AND FDP TENDONS

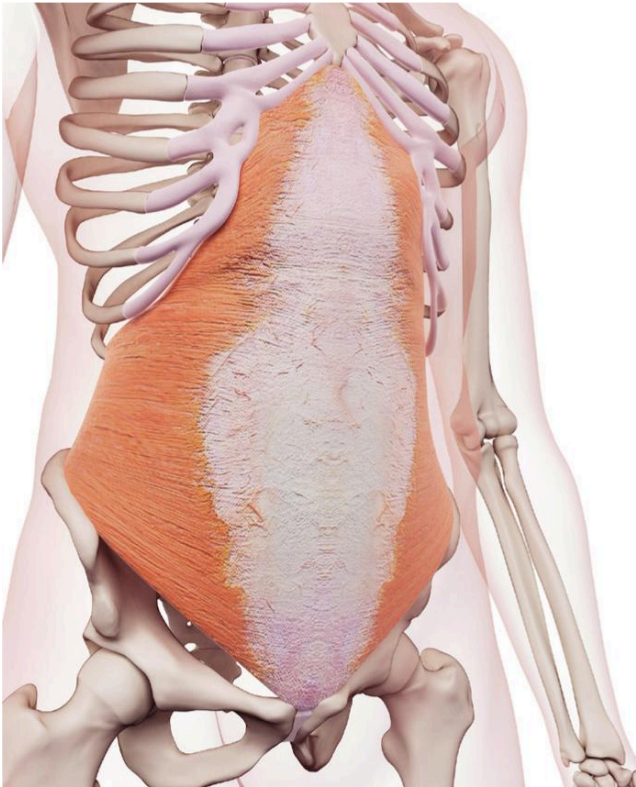
The hand is one of the most complex parts of the human body, biomechanically speaking, due to its broad range of movement and incredible fine motor control. Without detracting from this impressive complexity, we'll just focus on the finger tendons and ligaments (pulleys) since they are most prone to injury, as seen in the section on injury prevention.

CORE: THE CONNECTING CHAIN

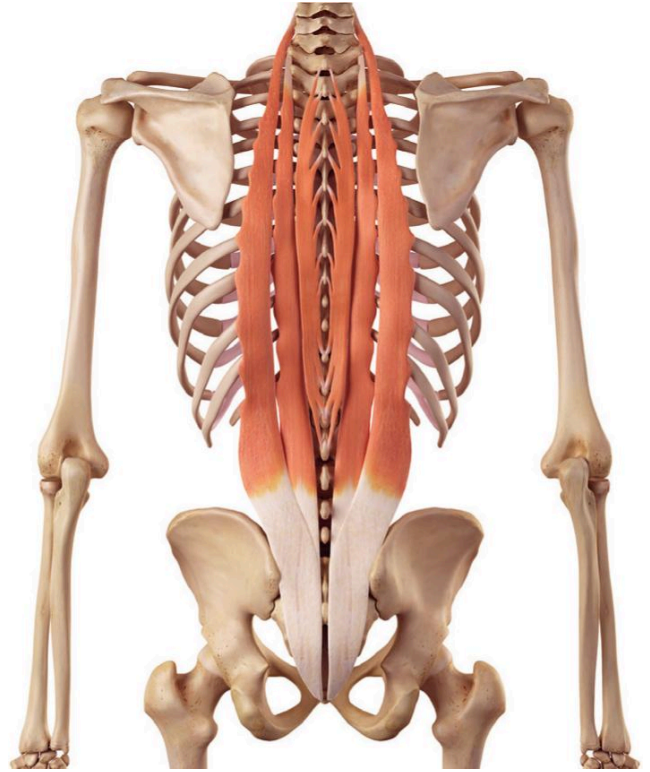
The core (centre or nucleus) has a dual function: it transfers force between the upper and the lower body, and it also stabilises and protects the spinal column (remember there's a complex system of nerves that branch off from the spinal canal and that an injury anywhere along the spine could have devastating consequences).

Considering these complex functions, it is no surprise that the core consists of much more than just the abdominal muscles (Boyle, 2017):

- *Scapular muscles*: the rhomboids and the serratus muscles connect the scapulae (and therefore the arms) to the rib cage. They transmit force from the arms to the body and vice versa.
- *Deep core*: focusing on the transverse abdominal muscle and the multifidus muscle. These muscles perform anticipatory postural adjustments, which means they're activated before the rest of the muscles involved in any given movement in order to protect and stabilise the spine.



TRANSVERSE ABDOMINAL MUSCLE



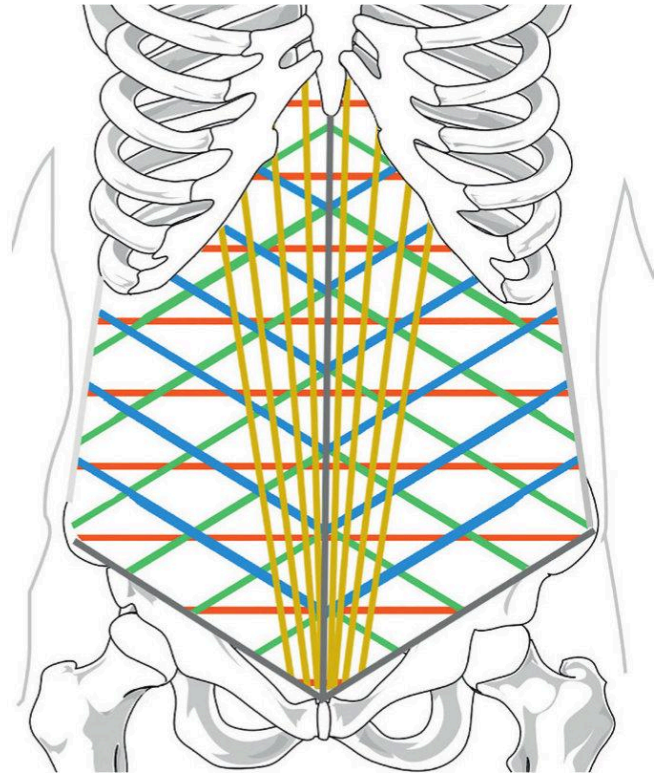
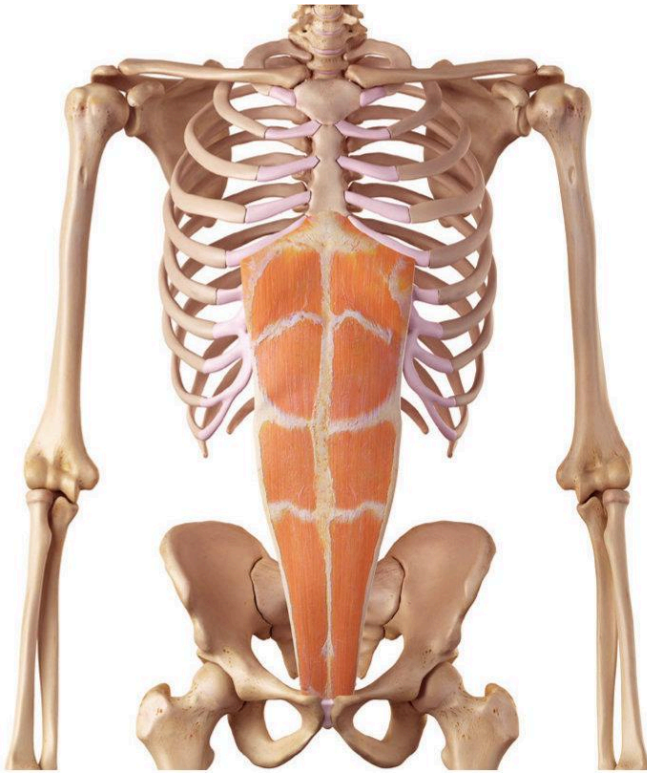
PARASPINAL MUSCLES



INTERNAL OBLIQUE (INSERTION IN THE FASCIA OF THE EXTERNAL OBLIQUE)

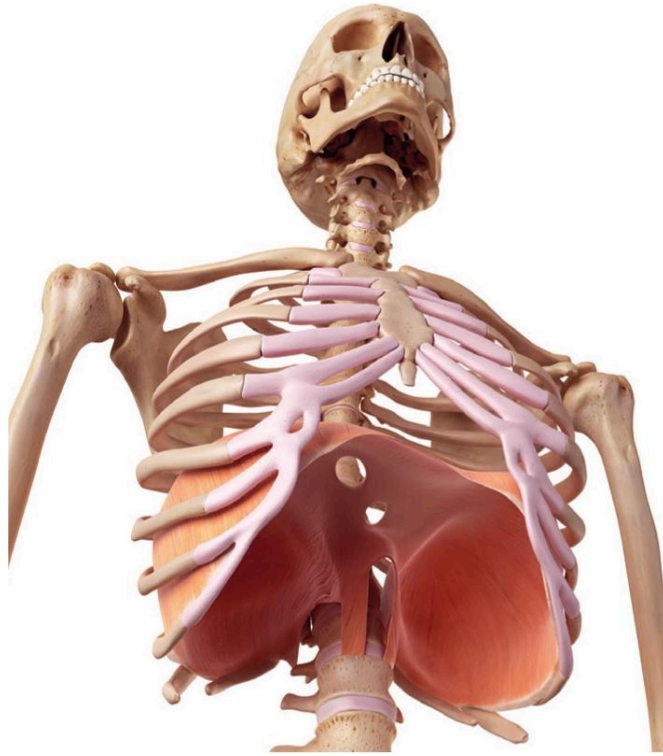


EXTERNAL OBLIQUE

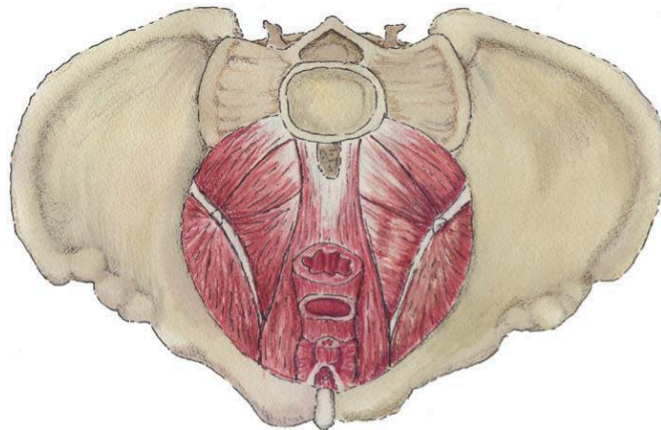


LEFT: RECTUS ABDOMINIS RIGHT: THE ABDOMINAL 'CORSET' (THE FIBRES OF THE TRANSVERSE ABDOMINAL MUSCLE ARE SHOWN IN ORANGE, THE FIBRES OF THE INTERNAL OBLIQUE IN GREEN, THE FIBRES OF THE EXTERNAL OBLIQUE IN BLUE AND THE FIBRES OF THE RECTUS ABDOMINIS IN YELLOW)

- *Abdomen muscles:* internal and external obliques, rectus abdominis and quadratus lumborum. This complex web of muscles forms a stabilising and protective corset around the lower back. There's a great deal of stress on this part of the body, so it's protected by having a very limited range of movement. The importance of the lower back for a long and healthy life cannot be overstated. The abdomen muscles also play a key role in holding the visceral organs in place. And, as we'll see, there are no 'upper', 'middle' or 'lower' abs. The rectus abdominis is made up of various muscle bellies, which form the different sections of the oh-so-coveted six-pack. It consists of one muscle on the left and one muscle on the right. And the way it looks has little to do with how much we train, as the size and layout of the muscle bellies is genetic.
- *Diaphragm and pelvic floor:* two large dome-like muscles that top and tail the core. The diaphragm is essential for breathing. The pelvic floor holds absolutely everything in place but with the slight 'issue' of having two 'holes' (genital and anal). It's often forgotten about, but poor function can lead to health problems (for example, incontinence or prolapse). What's more, although it may seem irrelevant to climbing performance, it can also lead to incorrect activation of the abdominal corset, resulting in a loss of strength through the central zone.

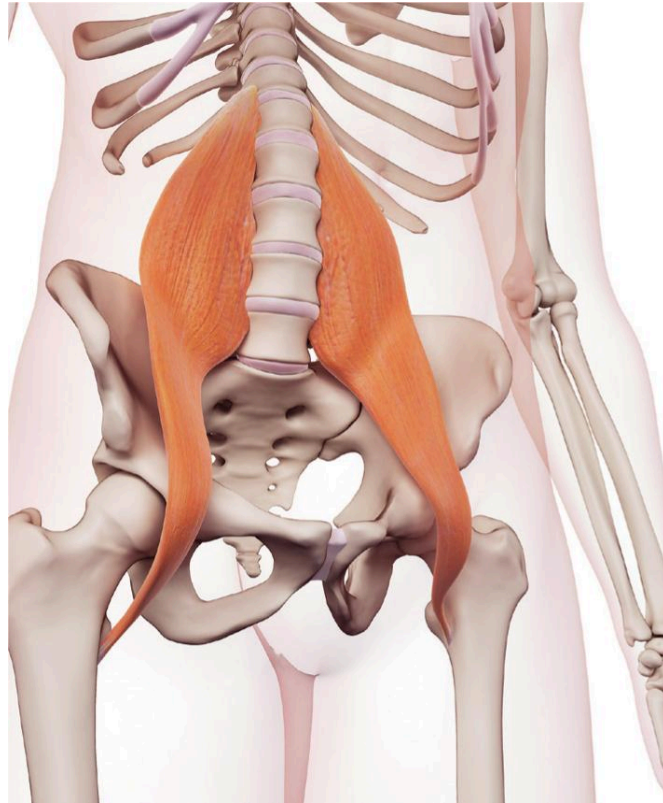


DIAPHRAGM



PELVIC FLOOR (PELVIS VIEWED FROM ABOVE)

- *Psoas*: the psoas connects the lower limbs to the lumbar spine and is one of the key muscles in hip flexion, meaning it helps you lift and place your feet when climbing. It's a really important muscle, both in terms of climbing movement and health of the lumbar region. If, for whatever reason, the psoas can't lift the femur, it will pull the lumbar region forwards. This exaggerates the natural curve in the lower back and puts pressure on the lumbar spine. The rectus abdominis stabilises the psoas and stops this from happening. The majority of supposed 'core' exercises, such as sit-ups or leg raises, are based on this biomechanical principle. They mainly work the iliopsoas muscle and, indirectly, the rectus abdominis as it tries to stabilise the iliopsoas. If the rectus abdominis isn't strong enough to do this, you'll end up with a nasty injury to the lumbar region.

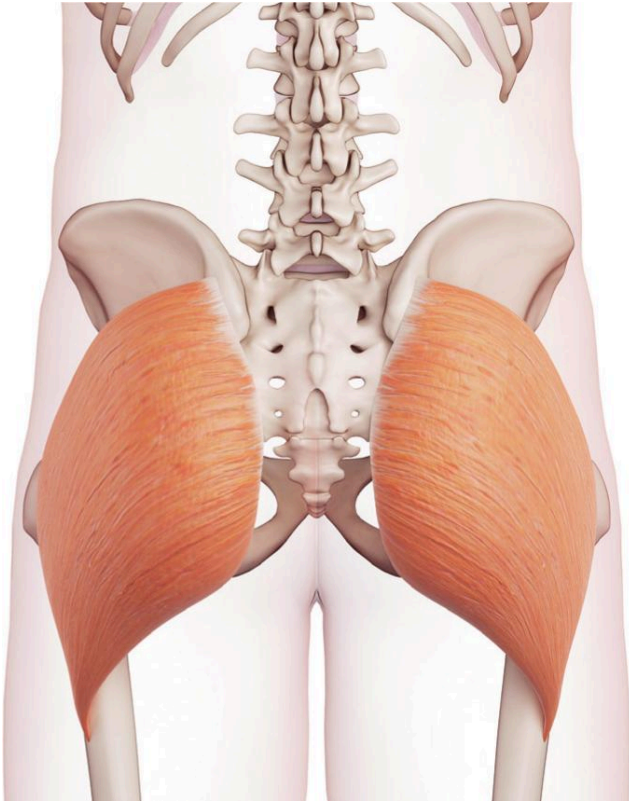


PSOAS

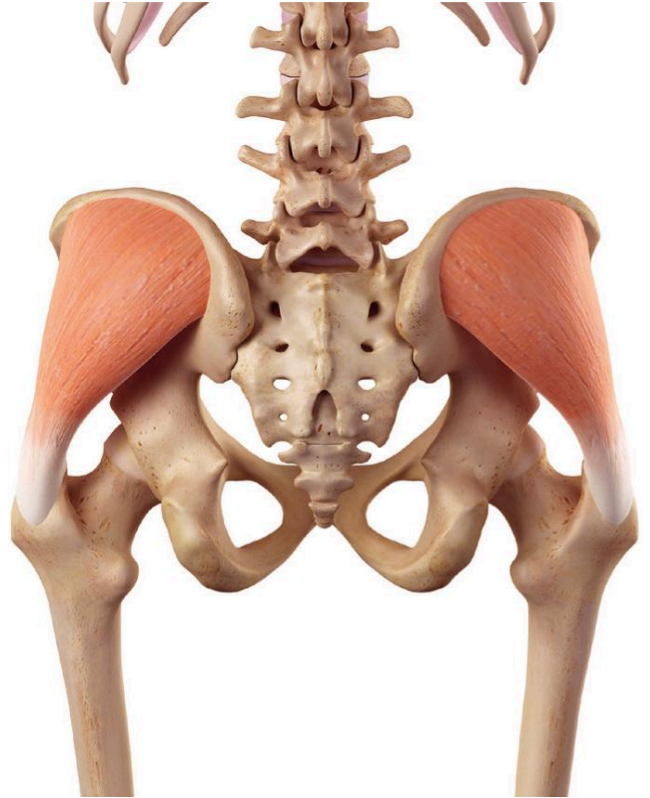
LOWER BODY: PUSHING MUSCLES

The lower body is the driving force behind both horizontal (floor-based) and vertical (wall-based) propulsion. It has three key joints: the coxofemoral joint (hip), the knee and the ankle.

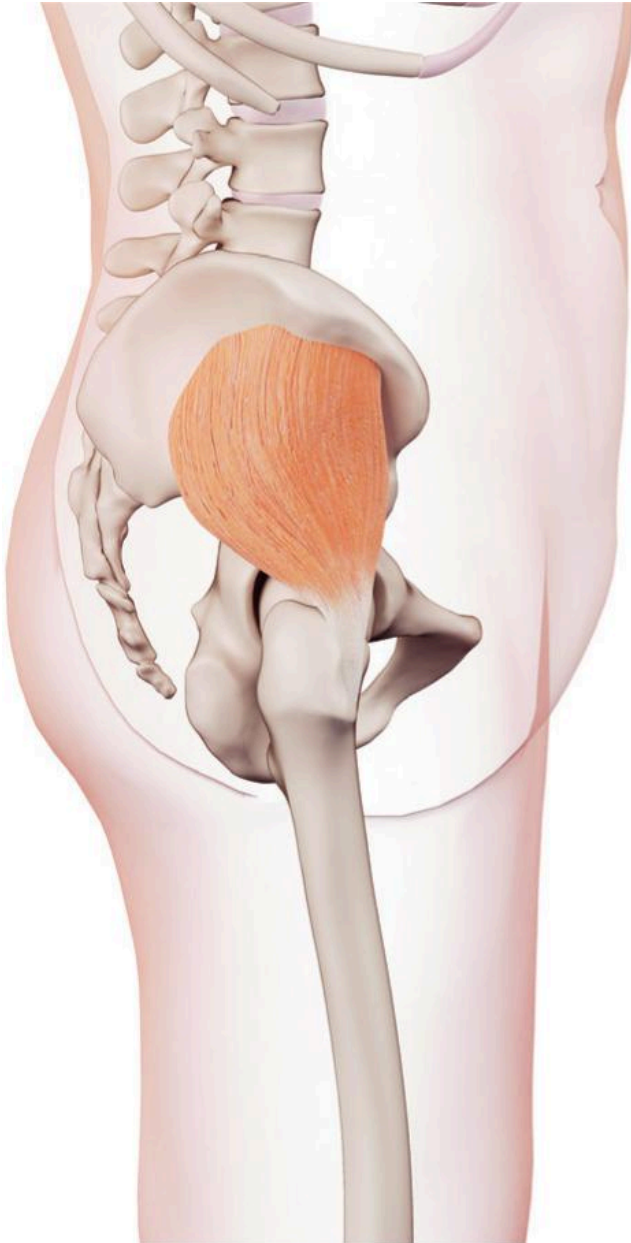
- *Hip muscles:* like the shoulder joint, the hip joint is amazingly complex, but it's less mobile and therefore much more stable. We'll focus here on the gluteus muscles around the hip joint. The gluteus maximus is the most powerful muscle in the human body and the primary hip extensor. When we've got a really high foot, it's what gives us the strength to weight it and stand up. The gluteus medius and minimus are mainly leg abductor muscles (they lift the leg away from the body), and they also help stabilise the hip joint. The gluteus maximus and the tensor fasciae latae, which is a small muscle at the front of the hip, mark the origin of the iliotibial band. This is a thick band of fascia that runs down the side of the thigh to below the knee, helping to transfer force to the lower leg. One of the heads of the quadriceps (the rectus femoris muscle) also originates in the pelvis, meaning that, in addition to its role in knee extension, it works in synergy with the psoas in hip flexion and leg elevation. This is why in exercises like sit-ups, we feel the burn in our quads (when we thought we were working our 'abs').



GLUTEUS MAXIMUS



GLUTEUS MEDIUS

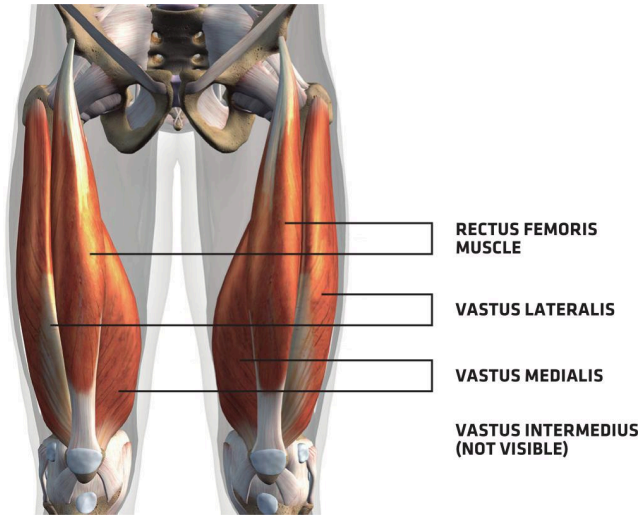


GLUTEUS MINIMUS



ILIOTIBIAL BAND (IN RED)

- *Thigh muscles:* to simplify things as much as possible, we'll only look at the quadriceps as a knee extensor and the biceps femoris as a flexor muscle (sending our apologies to the rest of the thigh muscles, like the abductors or the semimembranosus and the semitendinosus). Originating in the head of the fibula, the biceps femoris not only flexes but also externally rotates the knee. This is why you get a more powerful heel hook with your toes pointed outwards, but you could also end up damaging your knee if you pull too hard.
- *Lower leg muscles:* our focus here is on the tibialis anterior, which is located on the lateral side of the tibia and is the main muscle used for toe hooks, and the triceps surae, which is located on the back of the lower leg. The triceps surae consists of the gastrocnemius muscle (which lifts the ankle, allowing us to stand on our tiptoes, and is also responsible for knee flexion) and the soleus (which lifts the ankle when the knee is bent).



QUADRICEPS (NOTE THE ORIGIN OF THE RECTUS FEMORIS IN THE PELVIS, WHICH IS WHY IT CONTRIBUTES TO HIP FLEXION)



BICEPS FEMORIS (NOTE THE ORIGIN OF THE LONG HEAD IN THE PELVIS, WHICH IS WHY IT CONTRIBUTES TO HIP EXTENSION)



GASTROCNEMIUS

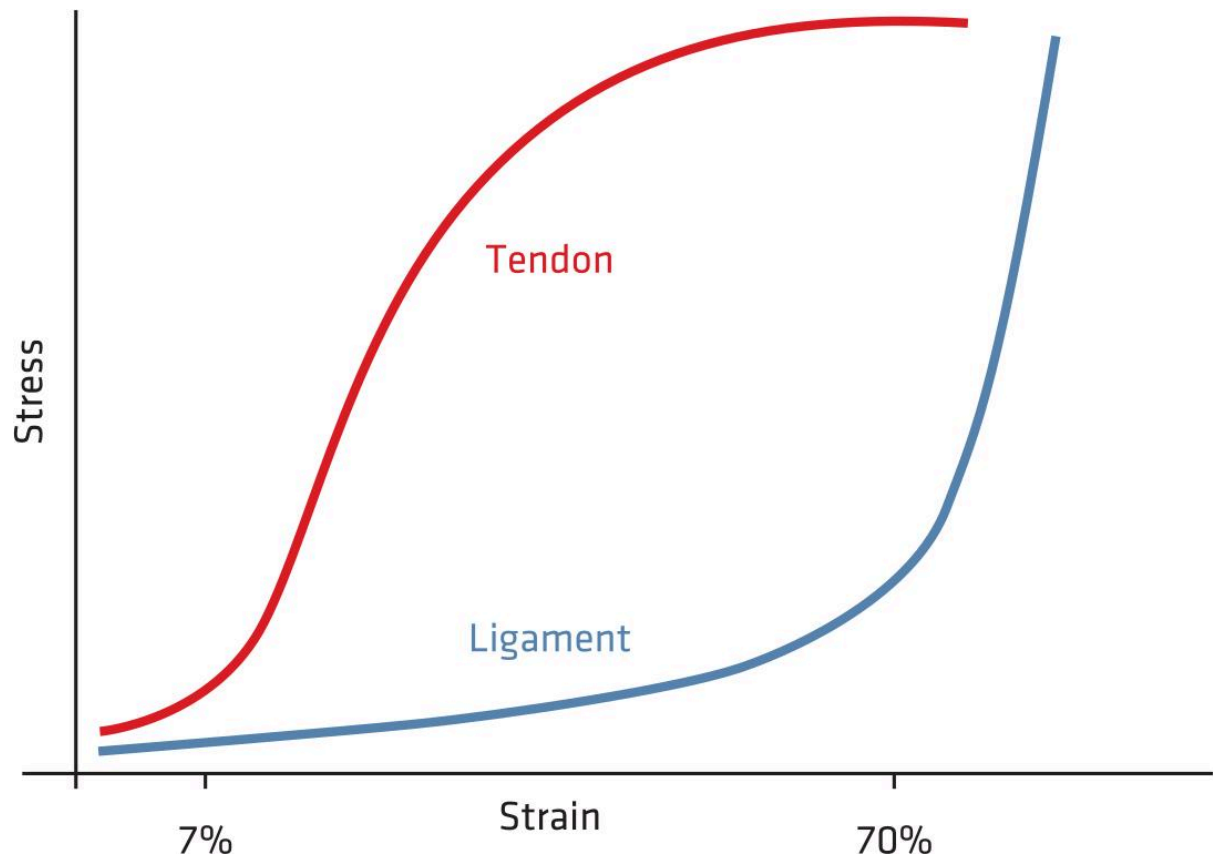


SOLEUS

CONNECTIVE TISSUE: TENDONS AND LIGAMENTS

Although tendons and ligaments are completely different, they are often confused and there is a general misunderstanding of what they are and how they work. A better understanding of tendons and ligaments will help us take better care of them and lower our risk of injury.

Tendons transmit force from muscle to bone, and vice versa. They're mainly composed of collagenous fibres (90 per cent), which are flexible (note: not elastic) and highly stretch-resistant. Tendons are a bit like a static or semi-static rope. Imagine your car is stuck in a ditch. What happens if you try to use a dynamic rope to pull it out? Being dynamic (elastic), the rope just absorbs all the energy of your tugging and pulling. Very little force is transmitted to the car and it doesn't budge. If you instead use a static or semi-static rope, which isn't dynamic but is still flexible because you take good care of it, much more force will be transmitted to the car and you'll be able to pull it out. Tendons work in much the same way. Aside from collagenous fibres, they also contain elastic fibres (0.5 per cent), mucopolysaccharides (0.5 per cent) and water (9 per cent).



STRESS STRAIN CURVE FOR TENDONS AND LIGAMENTS

As for ligaments, their main function is to reinforce our joints. They limit joint mobility to prevent dislocation. Although they aren't particularly elastic, they have a little more give than tendons.